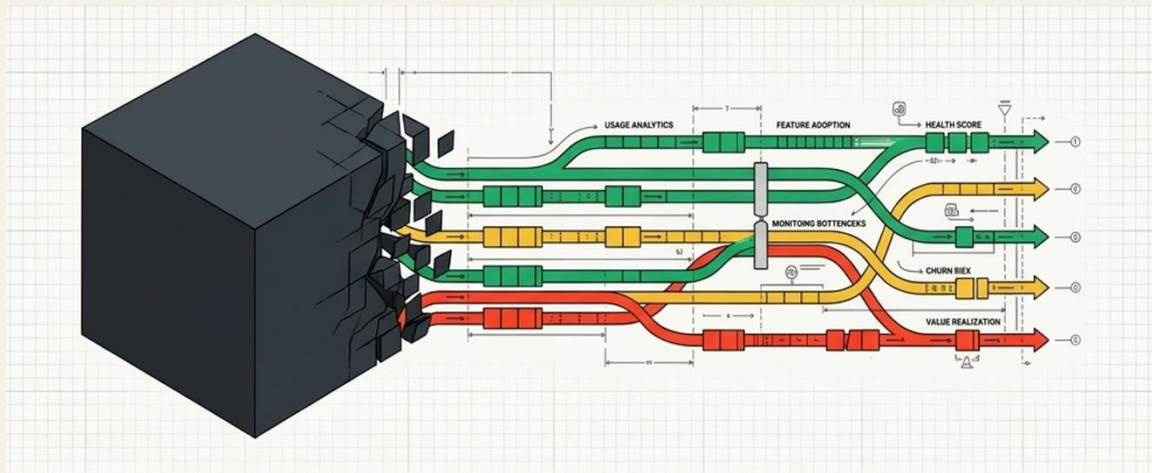


Proposal for Realized ARR: From Concept to Working Product

A North Star metric to move Global Customer Success
from static contract value to continuous customer value realization.

Agenda

- Introduction
- Context & Pain Points
- Proposal Metrics
- Technical Deep Dive
- Retrospective
- Future Roadmap
- The Ask



I build decision systems that help products scale



Customer & Business First

- Start with the Business decision and Customer outcomes.
- Built Azure CX decision systems that converted customer, support, incident, and usage signals into executive-ready business insights.



Design Thinking

- Start small, Scale fast, Adapt Better.
- Moved ambiguous Azure CX problems from concept to MVP across Service Health, quality reviews, AI summaries, and feedback workflows.



System Thinking

- Integrate people, product, process, tools and systems.
- Connected Product, Engineering, Support, CS, Field, and Data teams through shared operating rhythms, metrics, and governance.

I applied this exact philosophy to Realized ARR, moving from a metric concept to a fully functional BigQuery pipeline, automated tests, and a live executive dashboard.

Why are we here !

 [Live Demo](#)

My Recommendation

Approval: Approve a one-quarter shadow pilot in 1–2 regions before any compensation or operating model change.

Problem statement

GCS is moving from upfront Bookings / TCV toward an ARR + hybrid consumption model. Leadership needs a trusted way to measure customer value realization, guide post-sales action, and eventually inform incentives.

My thesis

GCS should not replace TCV; it should complement TCV with a shadow Realized ARR metric that identifies shelfware, recovery risk, and expansion readiness earlier.

In scope

Shadow operating metric
Customer value realization signal
Explainable MVP

Goal of Demo

Where should GCS recover ARR?
Where should GCS expand ARR?
What does good look like? What is missing?

Adoption strategy

This metric only works if paired with an operating rhythm: weekly CSM review, monthly regional review, quarterly weight calibration, and data-owner governance.

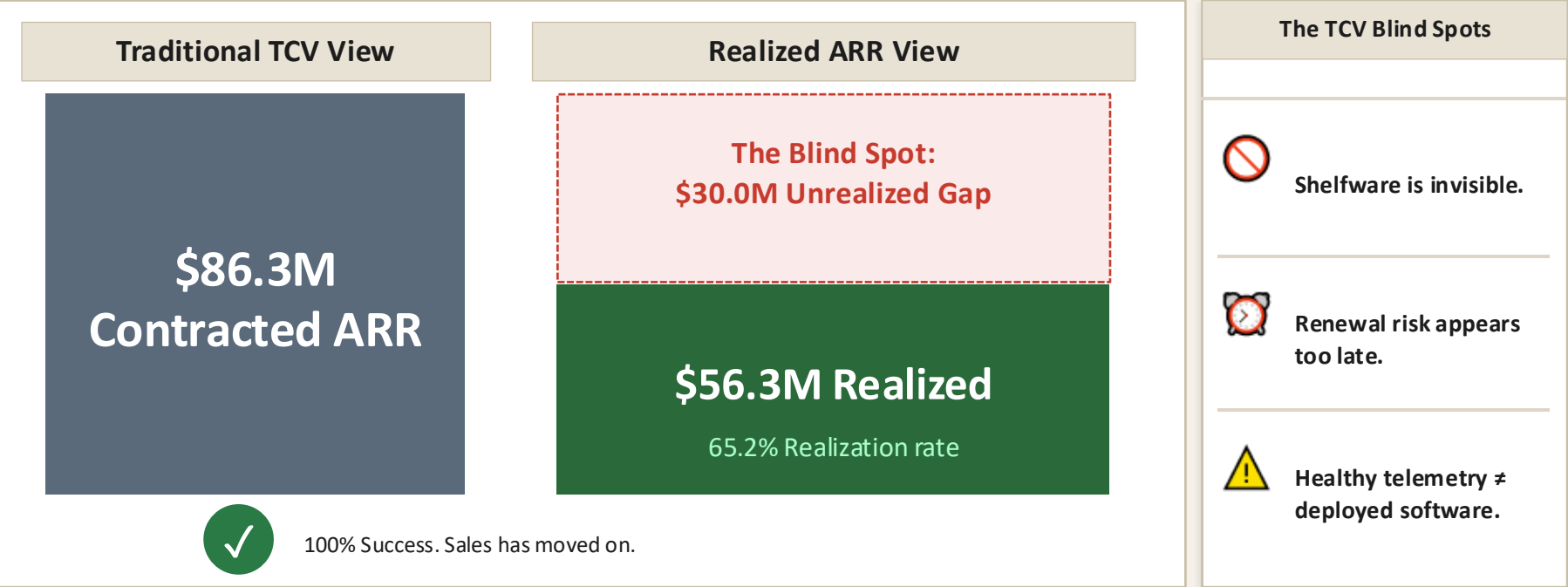
Out of scope

GAAP revenue recognition
Automated comp engine
Predictive black-box model

TCV hides the gap between contract value and customer value

The \$30M blind spot hidden in static contract values.

Dec 2024 Snapshot | 1,000 Accounts
12-month synthetic history



The business is actually won or lost in the 180 days after the contract is signed.

The Common Denominator Is Missing

Net Revenue Retention (NRR)

The Investor Standard



Sales / Execs | Post-sale

Flaw: Lags reality by 12 months. It is an outcome metric, not a behavioral leading indicator.

Customer Health Score (CHS)

The CS Standard



CSM | Reactive

Flaw: Highly subjective. Relies heavily on CSM input and NPS without a direct, objective tie-in to ARR.

Earned ARR (eARR)

The Predecessor



Product | Disconnected

Flaw: Incomplete logic. Fails to catch "spike-and-drop" consumption behaviors due to a lack of sustained usage dimensions.

Realized ARR

The GCS North Star



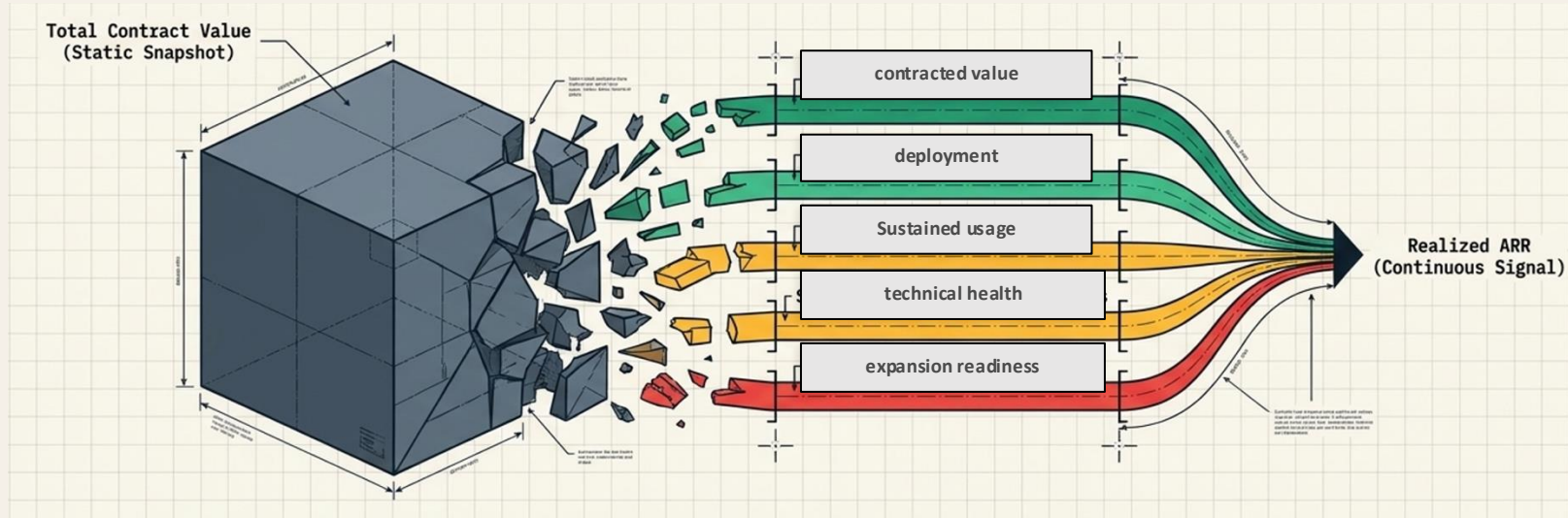
Win: Ties directly to ARR, surfaces shelfware early, handles spike-and-drop, and creates a stronger foundation for future incentive design.



Why Standard SaaS metrics fail to capture Phantom ARR

Signals Need to Be Plural, Not Singular

Proposal: Transitioning GCS from static signatures to continuous telemetry



GCS needs a decision metric that explicitly connects
contracted value → deployment → sustained usage → technical health → expansion readiness.

The Realized ARR Engine: Measuring consequence, not activity

$$PRS = (D \times 0.40) + (S \times 0.30) + (H \times 0.20) + (E \times 0.10)$$

Deployment (D) (40%)

Ingestion drives security efficacy.
Capped at 1.0 to prevent inflation.
MIN(1.0, consumed / included)

Expansion Momentum (E) (10%)

The forward-looking upsell flag.
Rewards consistent overagers.
Trailing 6-month usage trend

D 40%

S 30%

H 20%

E 10%

Technical Health (H) (20%)

Converts support signals into numeric values.
Missing records default to neutral (0.60).
AVG(Green=1.0, Yellow=0.6, Red=0.2)

Sustained Usage (S) (30%)

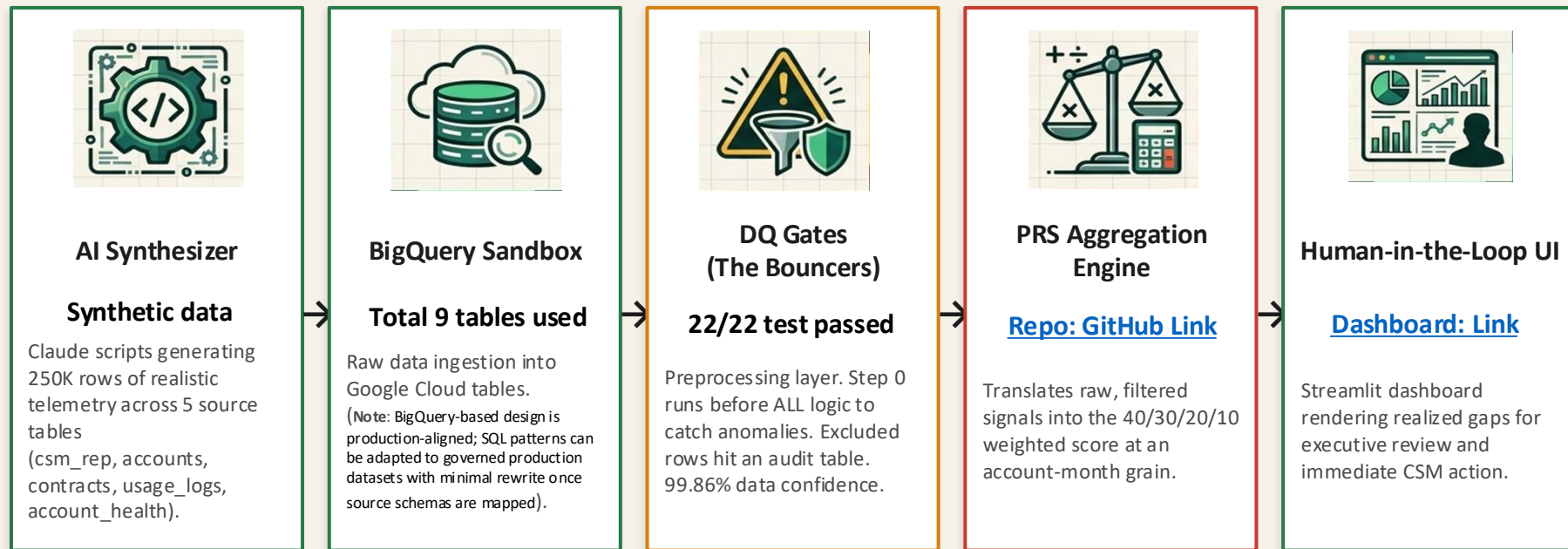
The spike-and-drop filter.
Uses a trailing 12-month cap.
healthy_months / 12-month window

The Shelfware Hard-Floor

Logic: If Deployment = 0 AND
Sustained Usage = 0, PRS is
hard-floored to 0.00.

Rationale: An undeployed product is an unprotected attack surface. It receives zero partial credit.

A production-aligned architecture built for governed data



I chose an explainable additive metric over a predictive ML model because the first trust barrier is organizational adoption, not model sophistication.

Designing for reality: Defending against data edge cases

Threat vs System Response

Rogue / Orphaned Logs

Usage disconnected from active accounts.

Hard Gates

Excluded before calculations. Cannot inflate deployment metrics. 99.86% confidence.

Test: test_s1_rogue___orphaned_log

Spike & Drop

Massive consumption Month 1, zero usage after.

12-Month Cap Logic

Sustained Usage component collapses account score to At-Risk immediately.

Test: test_s2_spike___drop

Mid-Year Expansions

Overlapping active contracts — double-counting risk.

Programmatic MAX() Logic

MAX(annual_commit_dollars) prevents ARR double-counting across contracts.

Test: test_s3_mid_year_expansions

Consistent Overagers

Sustained >120% consumption above included credits.

Upsell Flagging

Deployment capped at 1.0. Expansion Momentum flags 1.00 score for immediate upsell targeting.

Test: test_s4_consistent_overagers

Shelfware (D=0, S=0)

Deployed contract, zero usage, zero sustained activity.

Hard-Floor Override

PRS = 0.00. An undeployed product receives zero partial credit. No phantom value.

Test: test_s5_shelfware___d___s___

Translating business truth to prototype code via spec-driven AI

BRD → Realized ARR Spec v2 → Claude Code → pipeline → tests



1. Business Invariants First

Strip away SaaS assumptions. Define absolute truths of cybersecurity reality.

Example: Shelfware is a catastrophic security risk, not just a delayed budget realization.



2. Deterministic System Design

Translate truth into explicit data quality gates and mathematical edge cases.

Proof Point: 1,449 lines of rigorous Markdown specs (prd.md, sys_arch.md).



3. AI-Accelerated Execution

Leverage Claude Code to generate first-pass implementation, then constrain with specs, code review, and tests.

Proof Point: Automated generation of BigQuery SQL, Python, Streamlit UI, and 250K rows of highly realistic synthetic B2B SaaS telemetry.



[Live Demo](#)

\$30M unrealized. 100 shelfware accounts. H2 health declining. The evidence is live.

Synthetic December 2024 · 1,000 accounts · 12-month history · dbhagatnsdemo.streamlit.app

\$86.3M

Contracted ARR

\$56.3M

Realized ARR

\$30.0M

Unrealized gap

65.2%

Realization rate

H2 Portfolio Decline Story

Jan 2024	54.6%
Mar 2024	65.4%
Jun 2024 ▲ Peak	70.6%
Sep 2024	67.2%
Dec 2024 ▼	65.2%

Evaluation framework proof

22/22	tests passing
DQ rules · invariants · edge cases	
99.86%	data confidence
350 anomaly rows excluded	
5 types	B2B anomalies
All detected and handled	injected
Live	dashboard deployed
No setup · public URL	

→ **Live demo:** dbhagatnsdemo.streamlit.app · Home → By Region → By Rep → By Account → Recommended Next Action

Hard truth: This should not drive compensation on day one. Run as a shadow metric. Validate trust. Then scale.

If I Rebuilt This Tomorrow: Technical Retrospective

What I'd do differently today to scale from MVP to Enterprise Production



Architecture: Move to dbt

Raw SQL works for MVPs, but dbt scales. It provides superior lineage, incremental models, and built-in schema testing.



Validation: Empirical Weight Testing

The 40/30/20/10 split is business logic today. Before impacting compensation, weights must be tested empirically against historical renewal data from Day 1.



Data Sources: Integrate Real Support Data

The current technical health score is a proxy. We need real P1/P2 severities, error rates, and MTTR to make it defensible to SC.



Workflow Design: Dashboard for trust, push workflows for action

Dashboards create transparency, but behavior change improves when AI-generated briefs push prioritized next actions to CSMs with human approval.

LLM explanation layer

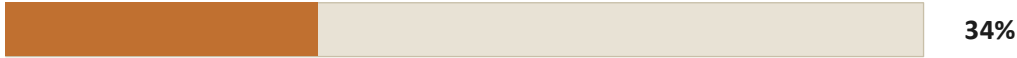
With today's Claude stack I'd build a narrative layer that explains PRS changes in plain English to CSMs the metric tells you the score, the LLM tells you why.

Human in the Loop:

AI surfaces the signal, humans validate the action

The Formula - What

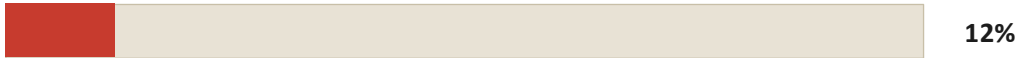
Account PRS: 0.34 (Orange/At-Risk)



Deployment: High (0.95)



Sustained Usage: Low/Dragging (0.12) - Critical Red Flag



Health: Yellow (0.60)



◆ The AI Narrative - Why

Account shows high initial ingestion, but zero telemetry for the last 14 days following a P1 ticket.

****Recommend dispatching PS support** to unblock data pipeline.**

Structured metrics quantify the exact gap. AI analytics translates fragmented signals into direct, actionable intelligence.

The Ask

Adoption Roadmap for shadow pilot

01 Approve Realized ARR as Shadow North Star metric

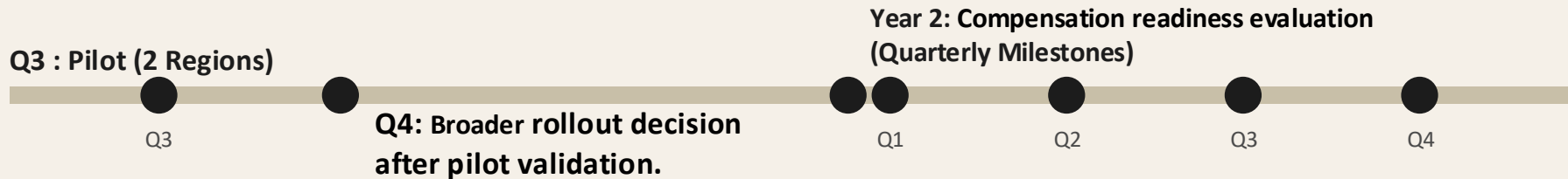
Run alongside TCV/ARR for one quarter.
Validate trust before compensation use.

02 Approve Controlled Pilot Timeline

Pilot 1–2 regions for one quarter with CS, SC, Support, Finance, and Data/AI.

03 Assign data owners for production signals

Unlock Customer Support Platform data.
Connect actual P1/P2 severities and MTTR to replace the health proxy.



Success criteria for Pilot

- Data confidence above agreed threshold | CSMs actively use weekly review
- Top recovery / expansion accounts are explainable
- Metric correlates with renewal, expansion, support, or deployment outcomes
- Exceptions are documented before any incentive discussion
- Compensation should be an output of validated behavior change, not the starting point.

Measure first. Validate behavior. Govern data. Then scale.
The metric is built. The tests pass. The dashboard is live.
The next step is a governed shadow pilot.

The Vision: Evolving to Agentic CSM Workflows & Adoption Roadmap

The Future of CS Is Outcome-Based

V1 Static Dashboard

Live Now

Realized ARR pipeline established. The metric tells you what is happening.



Agentic CSM Workflow (Phase 2 & 3)

AI continuously monitors accounts. It auto-generates weekly at-risk narratives and pushes briefs directly to CSM inboxes.

Human approval required. The AI recommends next action.
Dashboard remains the system of record; agentic briefs become the action layer.

LLM account narratives
CSMs validate accuracy / usefulness

Production signal expansion
Data confidence threshold agreed



Predictive Renewal Model (Phase 4 & 5)

6-month PRS trajectories predict renewal probability 180 days out, unlocking a new platform breadth dimension (rewarding multi-SKU deployments).

Comp readiness, back-test before incentive use.

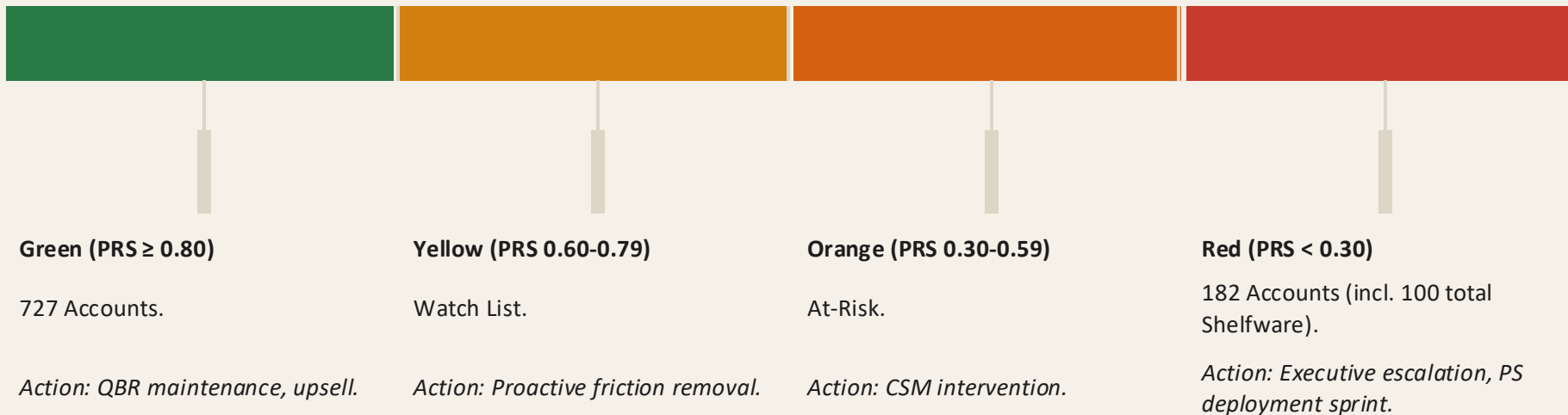
The metric tells you what. The AI tells you what to do. The CSM delivers the outcome. Neither replaces the other.

Q&A

Appendix

Portfolio Segmentation: Identifying where to deploy GCS capacity

Automated tests passing: 22/22



The synthetic portfolio PRS peaked at 0.706 in June and declined to 0.652 by December.

The Hard Truth: New ARR additions in H2 were mathematically less healthy than the existing base.

Every account gets a color, a dollar amount, and a recommended next action

Portfolio PRS bands replace subjective health scores with mathematically consistent, dollar-denominated urgency tiers.

Green PRS ≥ 0.80	727 accounts \$58.3M ARR	81.1%	→ QBR cadence maintenance · Identify upsell candidates · Expand platform breadth	Healthy
Yellow PRS 0.60–0.79	71 accounts \$4.0M ARR	10.3%	→ Proactive friction removal · Check deployment blockers · 30-day check-in call	Watch
Orange PRS 0.30–0.59	20 accounts \$1.2M ARR	1.7%	→ CSM-led intervention plan · PS deployment sprint · Escalate if no movement in 14 days	Intervene
Red PRS < 0.30	182 accounts \$1.8M ARR	8.7%	→ Executive escalation required · PS deployment sprint · 45-day recovery plan	Critical